Systems programming

Lec 4: Memory and Pointers

Somaia M. Elimam

جامعة الإمام المهدي
University of El Imam El Mahdi

Pointers and addresses

- A typical machine has an array of consecutively numbered or addressed memory cells.
- These memory cells can be manipulated individually or in contiguous groups
- A pointer is a group of cells that can hold an address.
- We use the * and & operators to deal with pointers.

```
int x = 1, y = 2, z[10];
int *ip;          /* ip is a pointer to int */

ip = &x;          /* ip now points to x */
y = *ip;          /* y is now 1 */
*ip = 0;          /* x is now 0 */
ip = &z[0];       /* ip now points to z[0] */
```

Pointers and function arguments

- In call by value, there is no way for the called function to alter variables in the calling function.

```
swap(a, b);

void swap(int x, int y)  /* WRONG */
{
     int temp;

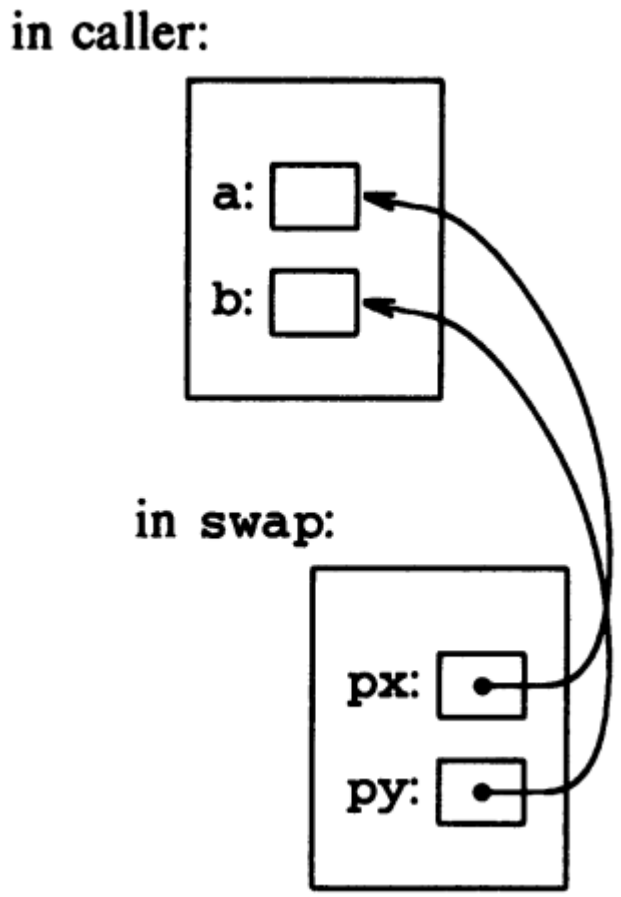
     temp = x;
     x = y;
     y = temp;
}
```

Pointers and function arguments

```
swap(&a, &b);


void swap(int *px, int *py)
{
     int temp;

     temp = *px;
     *px = *py;
     *py = temp;
}
```



Pointers and arrays

- Arrays and pointers are closely related in C.
- You can often think of an array variable as a pointer to the first array element, and a pointer variable as an array.
- However, they are not the same.

Arrays Are Not Pointers

```
char arr[] = "string";
char arr2[] = arr;
```

“error: invalid initializer”

```
char arr[] = "Hello World";
char *ptr = arr;
```

ptr points to arr[0].

Address arithmetic

- if p is a pointer to some element in an array, then:
  - P++ increment p to point to the next element in the array.
  - P+=i increment it to point to i element next to its current location.

Summary

- Memory locations are identified by addresses.
- Addresses are integers.
- Our system's memory is like one large array.
- POSIX processes appear to have their own dedicated memory.
- Pointers hold addresses and have types.
- Pointers and arrays are closely related, but not the same.

Required Readings

[1] Brian W. Kernighan and Dennis M. Ritchie. *The C Programming Language*. Second Edition. Chapter 5: Intro, 5.1-5.4. Prentice Hall, 1988.

References

- 2023 Ethan Blanton & Carl Alphonce / CSE 220: Systems Programming course